

Labelling less

People labelled the first 1,000 reviews. Active learning, rules and LLMs get the next 99,000 for a fraction of the cost.

THE QUESTION

Labelling 99,000 more reviews by hand costs Rs 24.75 lakh. How do you get usable labels while paying people for far fewer?

THE INTUITION

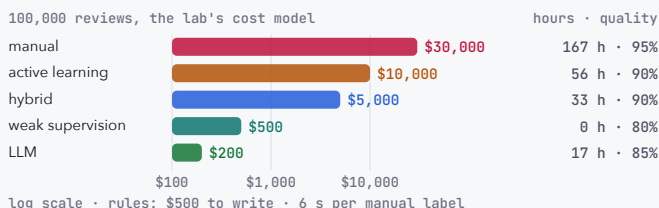
Spend human time only on items the model is unsure about. Let cheap sources, rules written as code or an LLM, label the rest, then look for labels that are probably wrong.

KEY TAKEAWAY

Every shortcut adds noise. Weight sources by accuracy, send unsure items to people, and spot-check a sample before trusting a batch.

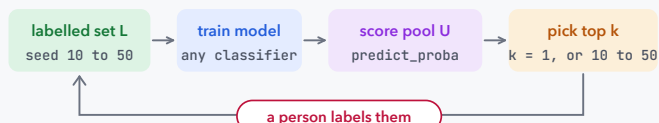
100,000 unlabelled → rules vote → LLM labels → model picks hard ones → people label those → flag mislabels → train

1 Labelling costs more than training



At Rs 25 and 5 minutes a review, 99,000 reviews take 8,250 hours; training takes hours. Claimed savings: active learning 2 to 10 times fewer labels, rules 10 to 100 times faster, LLMs 10 to 50 times cheaper.

2 Active learning lets the model choose



A review the model calls 99% positive teaches little; one near 50:50 sits on the decision boundary. modAL wraps the loop as `learner.query` and `learner.teach`.

```
for i in range(budget):
    probs = model.predict_proba(pool)
    q = (1 - probs.max(axis=1)).argmax()
    y_new = ask_human(pool[q])
    X_lab = np.vstack([X_lab, pool[q]])
    y_lab = np.append(y_lab, y_new)
    pool = np.delete(pool, q, axis=0)
    model.fit(X_lab, y_lab)
```

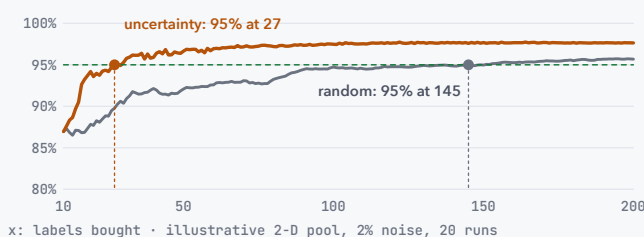
3 Three scores for "unsure"

review	P(pos, neu, neg)	1 - max	margin	entropy
Amazing film!		0.05	0.92	0.33
Terrible movie		0.05	0.92	0.33
It was okay		0.60	0.05	1.56
Interesting...		0.55	0.00	1.37
Not bad		0.52	0.00	1.20

shaded: what each score would query next · entropy in bits

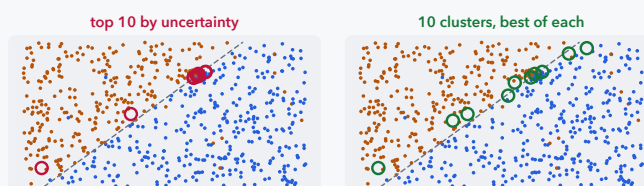
Least confident $1 - \max_k p_k$ and entropy $-\sum_k p_k \log p_k$: high is unsure. Margin $p_{(1)} - p_{(2)}$: small is unsure. With two classes all three rank items the same way; with three they can disagree.

4 Uncertain picks reach the target sooner



Same model and pool; only the choice of item differs. Uncertainty sampling reaches 95% with 27 labels, random sampling with 145, about 5 times more. Early uncertainty is unreliable (cold start): seed with 10 to 50 random or stratified items.

5 A batch needs diverse picks



40 near-duplicates: 7 of the top 10 in the clump, 3 of the clustered

Query 10 to 50 at a time. The top 10 often repeat one confusion, so cluster the uncertain items and take the best of each; add random picks so rare classes are not skipped. Stop when the budget runs out, accuracy plateaus, or all items left are over 95% confident.

6 Rules written as code vote or abstain

review	rating	1	2	3	4	5	6	7	8	vote	truth
amazing! Best film of the year	9.2	+			+	+				POS	POS
Terrible waste of time. Awful	2.1	-					-			NEG	NEG
okay, nothing special, not bad	5.5									none	NEG
Oscar-worthy... A masterpiece	8.8				+	+				POS	POS
fell asleep... disappointing	3.0								-	NEG	NEG
Great effects but weak story	6.5									none	POS
A delightful film... enjoy!	8.0									none	POS
The worst movie... garbage	1.5		-				-			NEG	NEG
Interesting but poorly executed	4.5									none	NEG
Five stars! Would recommend	9.0				+	+				POS	POS

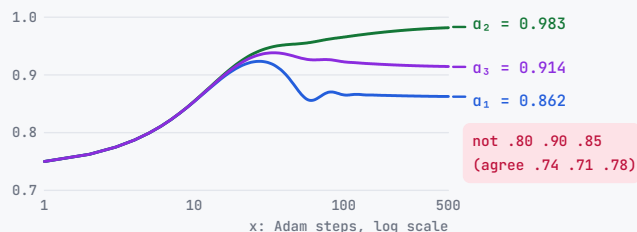
1 amazing 2 terrible|awful 3 worst|garbage 4 best|masterpiece
5 rating > 8 6 rating < 3 7 recommend 8 boring|disappointing

Each labelling function (LF) encodes one hunch. Here the 8 LFs never clash and are right whenever they fire, but the 4 hedged reviews get no vote. The lab's 8 LFs cover half its 818 reviews.

Weight each source by how often it is right

Estimate rule accuracy from agreement, weight the votes, check LLM labels, then hunt for the labels that are wrong.

7 Agreement reveals accuracy without labels



$$P_{ij} = \alpha_i \alpha_j + (1 - \alpha_i)(1 - \alpha_j) = \frac{1 + a_i a_j}{2}$$

with $a = 2\alpha - 1$. So $a_i a_j = 2P_{ij} - 1$, and agreement rates of 0.85, 0.80 and 0.90 fix all three: $a_1^2 = (a_1 a_2)(a_1 a_3)/(a_2 a_3)$. Gradient descent on the squared error agrees. N LFs give $N(N-1)/2$ equations for N unknowns: at least 3 LFs.

8 Log-odds weights settle conflicts

review	LF1 w 1.39	LF2 w 2.20		P(POS)
1 good movie (8.0)	POS	POS		97%
2 good but boring (5.0)	POS	abstain		80%
3 terrible (2.0)	abstain	NEG		10%
4 decent film (7.5)	abstain	POS		90%
5 good acting, bad plot	POS	NEG		31%
6 poor, great story	NEG	POS		69%

rows 5 and 6 tie under majority vote; the better LF wins

$$w = \ln \frac{\alpha}{1 - \alpha}, \quad P(\text{POS}) = \sigma \left(\sum_{\text{POS votes}} w - \sum_{\text{NEG votes}} w \right)$$

With α of 0.8 and 0.9 the weights are 1.39 and 2.20. Train the final classifier on these probabilistic labels so it also labels reviews no LF fires on.

```
L = PandasLFApplier(lfs).apply(df)
lm = LabelModel(cardinality=2)
lm.fit(L, n_epochs=500, seed=42)
probs = lm.predict_proba(L)
```

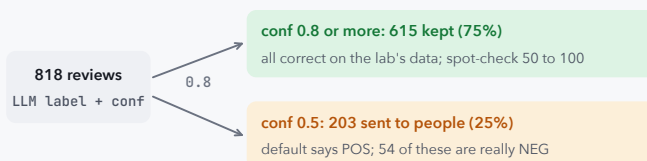
9 Pick by data size

under 1,000	label by hand
1,000 to 10,000	active learning, plus rules if patterns exist
over 10,000	rules or an LLM, plus active learning for hard cases

An LLM matches people on 72% of a sample?

Under 80%: fix the prompt before labelling the rest.

10 An LLM labels the bulk; people check it

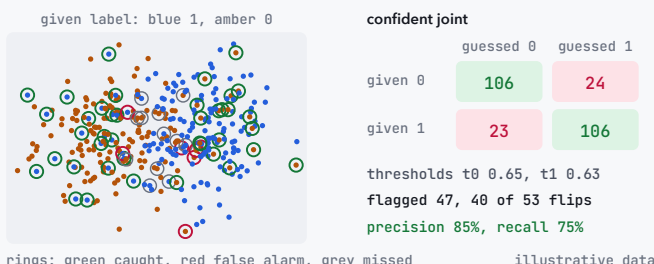


Put the task, label definitions and output format in the system message, one item per user message, temperature 0. Few-shot examples help. Per 1,000 labels the lecture lists \$300 to 500 for experts, \$20 to 50 for a large LLM, \$1 to 5 for a small one.

```
client.chat.completions.create(
    model=MODEL, temperature=0,
    max_tokens=10, messages=[
        {"role": "system", "content": RULES},
        {"role": "user",
         "content": f"Review: {review}"}])
```

Check 50 to 100 random labels against a person: over 90% agreement, keep the batch; under 80%, fix the prompt or use people. LLMs slip on irony ("so bad it's good" read as negative), specialist jargon, scales with many grades and vague guidelines.

11 Confident learning flags likely mislabels



Use out-of-fold probabilities. Class j 's threshold t_j is the mean p_j over items labelled j ; an item whose probability for another class clears that class's threshold is flagged. On the digits notebook (53 of 357 labels flipped) cleanlab flagged 56: 45 real, 11 false alarms, 8 flips missed.

```
pp = cross_val_predict(clf, X, y, cv=5,
                       method="predict_proba")
idx = find_label_issues(y, pp)
```

Remove, relabel or smooth the flagged labels. Under 5% noise is usually fine; over 15%, fix the labelling process.

people for the hard cases · rules and LLMs for the bulk · weight by accuracy · check a sample · flag mislabels

Source: lecture "Optimizing the labeling process", its demos (Snorkel labelling functions, LF accuracies from agreement, LLM labelling, cleanlab on digits) and its lab (synthetic reviews with seed 42, cost model), nipunbatra.github.io/stt-ai-teaching. The active learning, batch and confident-learning figures use illustrative data. How the first 1,000 labels were made and checked is on the labelling sheet.